Human-Centered Mobility Standards

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Purpose

These standards were developed to evaluate and guide the design of mobility systems, assistive technologies, vehicles, infrastructure, and other human-supporting environments from the perspective of human biological requirements.

They are based on the premise that human bodies should not be required to accommodate the limitations of machines, but that machines and environments should be designed to accommodate the needs of human bodies.

The standards emphasize mobility, standing, movement, anatomical accommodation, stability, shock reduction, and long-term physical well-being.

Foundational Principle

Human beings are biological organisms with physical structures, tissues, joints, organs, sensory systems, and movement requirements that evolved within the context of living bodies interacting with the physical world.

Any mobility device, vehicle, tool, building, surface, garment, workstation, or piece of infrastructure intended for human use should be designed to accommodate the needs of the human body rather than requiring the human body to adapt to the limitations of the machine or built environment.

The primary design objective should be the preservation and support of human well-being, comfort, function, stability, mobility, and long-term physical health.

Human-centered design requires consideration of the entire body, including:

- Feet
- Ankles
- Knees
- Hips
- Pelvis
- Spine
- Rib cage

- Shoulders
- Neck
- Head
- Sensory systems
- Organs
- Pelvic floor
- Genital structures
- Soft tissues
- Connective tissues
- Nervous system

No region of the body should be ignored merely because it has historically received less attention in engineering, transportation, architecture, manufacturing, or industrial design.

The machine should adapt to the body.

The body should not be required to adapt to the machine.

Standing Rather Than Sitting

Preferred mobility systems:

- Support standing.
- Support movement.
- Support weight shifting.
- Avoid prolonged sitting.

Mobility solutions that require sitting should generally not be considered unless specifically requested.

Assisted Standing Support

If a user is unable to stand independently, the appropriate response is not necessarily to require sitting.

Mobility systems should, where practical, support upright standing through external structural assistance that preserves the benefits of standing while reducing the physical demands placed upon the individual.

Such systems should:

- Support the pelvis.
- Support the rib cage.
- Assist with weight bearing.
- Promote spinal alignment.
- Reduce excessive loading on the lower extremities.
- Permit natural breathing and movement.
- Avoid concentrating support forces on a single anatomical region.
- Preserve dignity, autonomy, and mobility.
- Support movement rather than immobilization.
- Permit changes in position, weight shifting, reaching, turning, and locomotion to the greatest extent possible for the individual.
- Recognize that the human body is designed for movement and that support systems should facilitate movement regardless of a person's level of ability.

Support structures should be designed to distribute forces through robust anatomical regions capable of safely accepting load.

When standing assistance is required, support should preferentially be provided through the pelvis and rib cage rather than through methods that create excessive pressure on the neck, shoulders, arms, hands, knees, feet, genital structures, or other vulnerable tissues.

The inability to stand without assistance should not automatically be interpreted as a requirement to sit. Human-centered design should consider standing-support systems, weight-support systems, and other upright mobility solutions before defaulting to seated alternatives.

The objective of support is not merely to maintain a static position. The objective is to enable meaningful movement, participation, mobility, exploration, interaction, and engagement with the environment to the greatest extent possible.

Full Foot Support

The human foot is a complete anatomical structure and should be treated as such by any mobility device, vehicle, platform, tool, workstation, or built environment intended to support standing.

Mobility systems should:

- Fully support the entire foot.
- Avoid toe overhang.
- Avoid heel overhang.
- Distribute pressure evenly across the foot.
- Allow the foot to rest in a natural configuration.

- Accommodate the full dimensions of the user's foot.

Full-foot support is a minimum requirement for human accommodation.

A platform that cannot accommodate the entire foot cannot be considered fully adapted to the human body.

The adequacy of foot support should not be determined by the duration of use. A platform that is too small for the foot remains too small whether a person stands on it for a few seconds, a few minutes, or many hours.

Designers should not assume that partial support becomes acceptable because exposure time is short. Human-centered design begins with providing sufficient space for the complete anatomical structure being supported.

The machine should be sized to the foot rather than requiring the foot to conform to the dimensions of the machine.

Natural Stance Width

Mobility systems should:

- Permit a natural stance width.
- Accommodate hip-width or wider foot placement.
- Allow adjustment of stance width.
- Avoid forcing the feet into unnaturally narrow positions.

The preferred stance width should be determined by the body's requirements rather than the packaging requirements of the machine.

Parallel Foot Placement

Preferred mobility systems allow:

- Feet side-by-side.
- Symmetrical loading.
- Neutral pelvic orientation.
- Natural balance strategies.

Systems requiring a permanently staggered stance should be carefully evaluated for their long-term effects on the body.

Pelvic, Genital, and Soft-Tissue Accommodation

Human bodies include genital structures, pelvic soft tissues, and pelvic-floor systems that require appropriate accommodation by any physical environment, device, vehicle, workstation, garment, or piece of infrastructure intended for human use.

Designs should:

- Provide sufficient stance width for the user's anatomy.
- Avoid forcing the legs into positions that compress or crowd genital tissues.
- Avoid prolonged pressure on genital structures, pelvic soft tissues, or the pelvic floor.
- Permit natural positioning of the pelvis and lower body.
- Allow users to adjust stance and weight distribution during use.
- Recognize that genital comfort is a legitimate ergonomic consideration.
- Evaluate physical designs based on their effects on the entire body, including genital anatomy.

A mobility platform should be large enough to support the feet and preferred stance without requiring compression, folding, crowding, or distortion of the body's natural resting posture.

Joint Preservation

Mobility systems should:

- Reduce cumulative joint wear.
- Reduce unnecessary impact loading.
- Reduce vibration transmitted through the body.
- Minimize stress on feet, ankles, knees, hips, spine, and pelvic structures.

Shock Absorption

Preferred systems include:

- Suspension.
- Pneumatic tires.
- Air-compression technologies.
- Air-cushion technologies.

- Hover technologies that reduce or eliminate direct transmission of surface irregularities into the body.
- Flexible or shock-absorbing materials.
- Vibration-reducing interfaces.

Mobility systems should seek to reduce the transmission of impact, vibration, and abrupt changes in surface elevation into the human body.

Where feasible, technologies that create separation between the body and ground-borne vibration—including pneumatic systems, air-cushion systems, magnetic levitation systems, hover systems, or other future technologies—should be considered because they have the potential to reduce cumulative loading on joints, connective tissues, soft tissues, and the nervous system.

Rigid systems that transmit surface irregularities directly into the body should be avoided whenever practical.

Stability

Preferred systems provide:

- A broad support base.
- Predictable balance characteristics.
- Secure footing.
- Stability through design rather than constant physical compensation.

Human Scale Rather Than Portability

Portability is not a primary design objective.

If portability conflicts with:

- Foot support,
- Stance width,
- Stability,
- Suspension,
- Comfort,
- Joint preservation,

then the requirements of the human body should take precedence.

Evaluation Criteria

When evaluating a mobility device, ask:

1. Does it fully support the foot?
2. Does it permit a natural stance width?
3. Can the feet remain parallel?
4. Does it preserve pelvic alignment?
5. Does it accommodate genital and soft-tissue comfort?
6. Does it minimize vibration and impact?
7. Does it support standing rather than sitting?
8. If standing assistance is required, does it support the pelvis and rib cage while preserving movement?
9. Does it reduce cumulative joint stress?
10. Is stability achieved through design rather than physical compromise?
11. Does the device adapt to the body rather than requiring the body to adapt to the device?

A mobility device that satisfies these criteria is likely to provide a higher degree of human-centered biomechanical support than one designed primarily around portability, compactness, or manufacturing convenience.